10 Robert makes 50 litres of green paint by mixing litres of yellow paint and litres of blue paint in the ratio 2 : 3

Yellow paint is sold in 5 litre tins.
Each tin of yellow paint costs £26

Blue paint is sold in 10 litre tins.
Each tin of blue paint costs £48

Robert sells all the green paint he makes in 10 litre tins.
He sells each tin of green paint for £66.96

Work out Robert's percentage profit on each tin of green paint he sells.

	Yellow	Blue	Total	
	2	3	5	
×10	20	30	50	×10

1 mark
20 L yellow
30 L blue

Yellow: 20 ÷ 5 = 4 tins, costing 4 × £26 = £104

Blue: 30 ÷ 10 = 3 tins, costing 3 × £48 = £144

Total cost = 104 + 144 = £248

1 mark
Cost the 50 litres

	Litres	GBP	
	50	248	
÷5	10	49.60	÷5

1 mark
One 10 L tin costs £49.60

Profit per tin = £66.96 – £49.60 = £17.36

	GBP	Percent	
	49.60	100	
÷100	0.496	1	÷100
×35	17.36	35	×35

1 mark
Compare profit with cost as a percentage

1 mark
35% profit

35 ..... %

(Total for Question 10 is 5 marks)